

Aishwarya Vikas Patil

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Technical Skills

- **Programming:** Python, JavaScript, SQL, R, HTML/CSS
- **Big Data Technologies:** Airflow, Spark, Kafka, Snowflake, Pinecone, MS SQL Server, PostgreSQL, Tableau, Power BI, MongoDB
- **Cloud & DevOps:** Databricks, AWS (S3, Redshift, Lambda), Google Cloud Platform (GCP), Microsoft Azure (SQL, Data Factory), Docker, Kubernetes, Git/GitHub
- **ML Libraries & AI:** Pandas, NumPy, PyTorch, scikit-learn, Matplotlib, OpenCV, LLM (Anthropic, OpenAI, Gemini), RAG, LangChain, Guardrails, Embeddings

Work Experience

Mundoprints, AI/Data Intern, Boston, MA

Feb 2025 - Jun 2025

- Automated internal **ETL** pipelines in **Python** + **MSSQL** using parameterized CTE logic and scheduled automation, reducing manual data quality check time by **70%**.
- Built a competitor pricing scraper using **Selenium** and **Playwright** to extract data from dynamic JS-rendered pages across **3** websites, processing **1,000+** product attribute updates weekly with normalization and deduplication; scheduled via **Airflow** and containerized with **Docker** for consistent dev-to-production deployment.
- Explored AWS data warehouse technologies including **S3** and **Redshift** as alternatives to PostgreSQL for scalable data storage and ingestion.
- Built a GenAI chatbot powered by **Claude API**, integrating pgvector with **PostgreSQL** for semantic search over customer query data.

Northeastern University, Teaching Assistant, Advance Data Science/Architecture, Boston, MA

Sep 2025 - Dec 2025

- Supported instruction for a graduate-level course covering **Neural Modeling, LLMs, and Causal Inference**, helping students work through concepts using Jupyter Notebooks.
- Held office hours and coding labs for **30+** students working in Pandas and R, providing hands-on debugging help and feedback on project submissions.
- Led weekly troubleshooting sessions focused on hierarchical temporal algorithms (HTA), breaking down complex logic to help students apply it in their assignments.

Projects

AutoInsight - Automotive Intelligent Analytics Platform | [Project Link](#)

Jan 2026

- Built a **multi-agent LLM system** (LangGraph + Claude SDK) with **RAG-powered schema retrieval** via Pinecone, achieving **95.3% SQL accuracy** and **87% token reduction** through multi-agent validation.
- Ingested **84GB** of automotive data (NHTSA, EPA, FRED) via REST APIs; ran **Apache Spark ETL** to load **224K+ records** into Snowflake.
- Enforced **RBAC guardrails** (Analyst, Manager, Executive) with column-level filtering, constraining SQL generation within validated vector store schema context.
- Deployed **FastAPI** + **Streamlit** on **AWS EC2** with sentiment analysis across **210K+ complaints** and daily orchestration via **Airflow/GCP**.

Food Inspection Analytics Platform | [Project Link](#)

Dec 2025

- Unified **450K+ records** from Chicago and Dallas using Alteryx with RegEx-based standardization of zip codes and risk categories.
- Architected **Delta Live Tables** on Databricks with **Medallion Architecture** (Bronze/Silver/Gold) handling schema evolution.
- Enforced **8 validation rules** in the Silver layer, eliminating **98% of entry errors** for high-integrity reporting.
- Designed a **star schema** with **SCD Type 2** and built Tableau dashboards to visualize violation patterns and compliance metrics.

Chinook Data Warehouse Pipeline | [Project Link](#)

Oct 2025

- Modeled a **dimensional data warehouse** in Snowflake (Kimball, 5+ fact/dimension tables) for media store analytics.
- Orchestrated end-to-end **ETL** pipelines via **Azure Data Factory** to extract from SQL Server, stage in **Azure Blob Storage**, and load into Snowflake
- Secured infrastructure using **Azure Key Vault** and **Managed Identity** for credential and storage access.
- Implemented **MD5 hash-based CDC** in ADF Data Flows for efficient incremental loading.

TMNIST Character Classifier | [Project Link](#)

Mar 2025

- Built a **CNN** (TensorFlow/Keras) to classify **94 characters** across font styles on TMNIST (**500K+ samples**), achieving **93.51% test accuracy** in 10 epochs.
- Designed a multi-layer architecture with **Conv2D, MaxPooling, and Dropout** to extract spatial features and prevent overfitting.

Education

Northeastern University, Boston, MA | *Master of Science, Information Systems*

Dec 2025

University of Mumbai, Mumbai, India | *Bachelor of Engineering, Computer Engineering*

Jun 2022